
CHAPTER 2

Numerical representation of data

When summarizing and describing numerical variables you need to do more than just prepare the tables and charts discussed in the last Chapter. In reading this chapter, you can learn some of the descriptive measures that identify the properties of central tendency, variation, and shape.

2.1 MEASURES OF LOCATION

These are measures of the centre of a distribution. They are single values that give a description of the data. They are also referred to as measure of central tendency. Some of them are arithmetic mean, geometric mean, harmonic mean, mode, and median.

2.1.1 THE ARITHMETIC MEAN (A.M)

The arithmetic mean (average) of set of observation is the sum of the observation divided by the number of observation. Given a set of a numbers $x_1, x_2, \dots, x_n$, the arithmetic mean denoted by $\bar{X}$ is defined by

$$\bar{X} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

Example. The ages of ten students are 16,20,19,21,18,20,17,22,20,17. The mean age is

$$\bar{X} = \frac{1}{10} \sum_{i=1}^{10} x_i = \frac{16 + 20 + 19 + 21 + 18 + 20 + 17 + 22 + 20 + 17}{10} = 19$$

If the values $x_1, x_2, \dots, x_n$ occur $f_1, f_2, \dots, f_n$ times respectively, then

$$\bar{X} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{n} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$$

Example. Find the mean for the table below

Scores (x_i)	2	5	6	8
Frequency (f_i)	1	3	4	2

The average value is

$$\bar{X} = \frac{\sum_{i=1}^4 f_i x_i}{\sum_{i=1}^4 f_i} = \frac{1 \times 2 + 3 \times 5 + 4 \times 6 + 2 \times 8}{1 + 3 + 4 + 2} = \frac{57}{10} = 5.7$$

Calculation of mean from grouped data

If the items of a frequency distribution are classified in intervals, we make the assumption that every item in an interval has the mid-values of the interval and we use this midpoint for x_i .

Example. The table below shows the distribution of the waiting time for some customers in a certain petrol station.

Waiting time (in min)	1.5 – 1.9	2.0 – 2.4	2.5 – 2.9	3.0 – 3.4	3.5 – 3.9	4.0 – 4.4	Σ
No. of customers	3	10	18	10	7	2	50
mid-values	1.7	2.2	2.7	3.2	3.7	4.2	
$f_i \times x_i$	5.1	22	48.6	32	25.9	8.4	142

The average waiting time of the customers is $\bar{X} = \frac{\sum_{i=1}^6 f_i x_i}{\sum_{i=1}^6 f_i} = \frac{142}{50} = 2.84$.

ADVANTAGE OF MEAN

The mean is an average that considers all the observations in the data set. It is single and easy to compute and it is the most widely used average.

DISADVANTAGE OF MEAN

Its value is greatly affected by the extremely too large or too small observation.

2.1.2 THE HARMONIC MEAN (H.M)

The H.M of a set of numbers $x_1, x_2, \dots, x_n$ is the reciprocal of the arithmetic mean of the reciprocals of the numbers. It is used when dealing with the rates of the type per (such as kilometers per hour, Dinar per liter). The formula is expressed thus

$$HM = \frac{1}{\frac{1}{n} \sum_{i=1}^n \frac{1}{x_i}} = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

If the values $x_1, x_2, \dots, x_n$ occur $f_1, f_2, \dots, f_n$ times respectively, then

$$HM = \frac{\sum_{i=1}^n f_i}{\sum_{i=1}^n \frac{f_i}{x_i}}$$

Example. The harmonic mean of 2,4,8,11,4 is $HM = \frac{5}{\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{11} + \frac{1}{4}} = \frac{440}{107} = 4.1121$

The harmonic mean takes into account every value and extreme values have least effect.

The formula breaks down when "0" is one of the observations.

2.1.3 THE GEOMETRIC MEAN(G.M)

The geometric mean is useful in finding the average change of percentages, ratios, indexes, or growth rates over time. It has a wide application in business and economics because we are often interested in finding the percentage changes in sales, salaries, or economic figures, such as the gross domestic product, which compound or build on each other. The geometric mean of a set of n positive numbers $x_1, x_2, \dots, x_n$ is defined as the n^{th} root of the product of n values. The formula for the geometric mean is written

$$GM = \sqrt[n]{x_1 \times x_2 \times \dots \times x_n}$$

If f_i is the frequency of x_i , then

$$GM = \sqrt[\sum_{i=1}^n f_i]{x_1^{f_1} \times x_2^{f_2} \times \dots \times x_n^{f_n}}$$

Example. The geometric average of 5, 8, 12, 25 and 34 is

$$GM = \sqrt[5]{5 \times 8 \times 12 \times 25 \times 34} = 13.247$$

Note that GM calculate takes into account every value but It cannot be computed when "0" is on of the observation.

Relation between Arithmetic mean, Geometric and Harmonic

In general, the geometric mean for a set of data is always less than or equal to the corresponding arithmetic mean but greater than or equal to the harmonic mean.

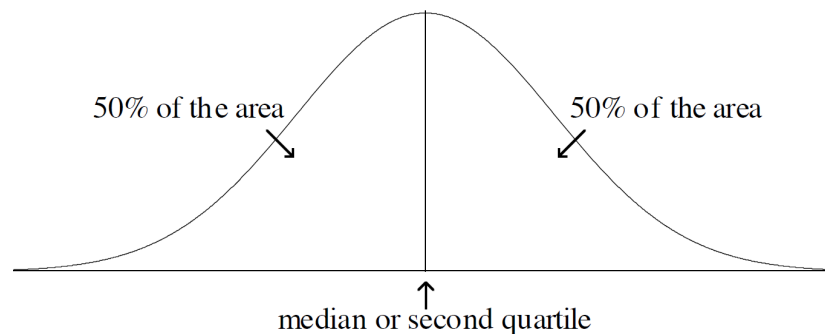
That is,

$$HM \leq GM \leq AM$$

The equality signs hold only if all the observations are identical.

2.1.4 THE MEDIAN

This is the value of the variable that divides a distribution into two equal parts when the values are arranged in order of magnitude.



If there are n (odd) observation, the median $\tilde{X}$ is the center of observation in the ordered list. The location of the median is $\left(\frac{n+1}{2}\right)^{th}$ item.

But if n is even, the median is the average of the two middle observations in the ordered list. i.e.

$$\tilde{X} = \frac{1}{2} \left(x_{(\frac{n}{2})^{th}} + x_{(\frac{n}{2}+1)^{th}} \right)$$

Example. The values of a random variable x are given as 8,5,9,12,10,6 and 4. Find the median ?

In an sorted array: 4,5,6,8,9,10,12. $n = 7$ is odd, therefore, the median is

$$\tilde{X} = x_{(\frac{7+1}{2})^{th}} = x_{(4)^{th}} = 8$$

Example. The value of a random variable are given as 15,15,17,19,21,22,25 and 28. Since $n = 8$ is even, the median is given by

$$\tilde{X} = \frac{1}{2} \left(x_{(\frac{8}{2})^{th}} + x_{(\frac{8}{2}+1)^{th}} \right) = \frac{1}{2} \left(x_{(4)^{th}} + x_{(5)^{th}} \right) = \frac{1}{2} (19 + 21) = 20.$$

Calculation of Median from a grouped data

The formula for calculating the median from grouped data is defined as

$$\tilde{X} = L_1 + \left(\frac{\frac{n}{2} - Cf_b}{f_m} \right) \times w$$

Where

L_1 =Lower class boundary of the median class

Cf_b =Cumulative frequency before the median class

f_m =Frequency of the median class

w =Class size or width

Example. The table below shows the height of 70 men randomly selected.

Height	118-126	127-135	136-144	145-153	154-162	163-171	172-180
f_i	8	10	14	18	9	7	4

Compute the median?

Height	Frequency	Cumulative frequency
118 – 126	8	8
127 – 135	10	18
136 – 144	14	32
145 – 153	18	50
154 – 162	9	59
163 – 171	7	66
172 – 180	4	70

We have $\frac{n}{2} = 35$. The sum of first three classes frequency is 32 which therefore means that the median lies in the fourth class and this is the median class. Then

$$L_1 = 145$$

$$Cf_b = 32$$

$$f_m = 18$$

$$w = 8$$

$$\tilde{X} = L_1 + \left(\frac{\frac{n}{2} - Cf_b}{f_m} \right) \times w = 145 + \left(\frac{\frac{70}{2} - 32}{18} \right) \times 8 = 146.33$$

The advantage of the median is that Its value is not affected by extreme values; thus it is a resistant measure of central tendency and It is a good measure of location in a skewed distribution, however, It does not take into consideration all the value of the variable.

2.1.5 THE MODE

The mode is the value of the data which occurs most frequently. A set of data may have no, one, two or more modes. A distribution is said to be uni-modal, bimodal and multimodal if it has one, two and more than two modes respectively. As example, the mode of scores 2, 5, 2, 6, 7 is 2.

Calculation of mode from grouped data

From a grouped frequency distribution, the mode can be obtained from the formula

$$\hat{X} = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \times w$$

Where

L_{mo} = lower class boundary of the modal class

Δ_1 = Difference between the frequency of the modal class and the class before it

Δ_2 = Difference between the frequency of the modal class and the class after it

w = Class size

Example. For the table below, find the mode.

Class	11 – 20	21 – 30	31 – 40	41 – 50	51 – 60	61 – 70
frequency	6	20	12	10	9	9

The modal class is the second class with $f_2 = 20$.

$$L_{mo} = 21$$

$$\Delta_1 = 14$$

$$\Delta_2 = 8$$

$$w = 9$$

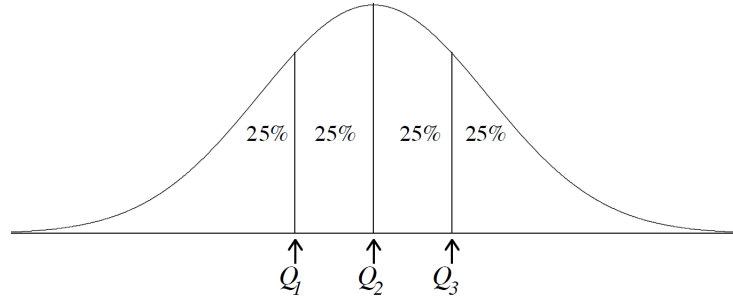
$$\text{hence, } \hat{X} = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \times w = 21 + \left(\frac{14}{14+8} \right) \times 9 = 26.727.$$

2.2 MEASURES OF PARTITION

From the previous section, we've seen that the median is the value that divides a distribution into two equal parts. Also there are other quantity that divides a set of data (in an array) into different equal parts. Such data must have been arranged in order of magnitude. Some of the partition values are: the quartile, deciles and percentiles.

2.2.1 THE QUARTILES

Quartiles divide a set of data in an array into four equal parts.



For ungrouped data, the distribution is first arranged in ascending order of magnitude.

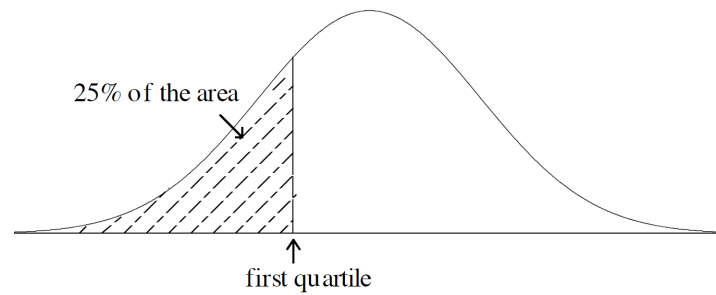
Then

First Quartiles : $Q_1 = \left(\frac{n+1}{4}\right)^{th}$

Second Quartile : $Q_2 = 2 \times \left(\frac{n+1}{4}\right)^{th} = \text{median}$

Third Quartile : $Q_3 = 3 \times \left(\frac{n+1}{4}\right)^{th}$ member of the distribution

if for $i = 1, 2, 3$; $i \times \left(\frac{n+1}{4}\right) \notin \mathbb{N}$, then $Q_i = \frac{x_{(k)} + x_{(k+1)}}{2}$ where k is the first integer before $i \times \left(\frac{n+1}{4}\right)$, $x_{(k)}$ and $x_{(k+1)}$ are the $(k)^{th}$ and $(k+1)^{th}$ ordered observations.



For a grouped data

$$Q_i = L_{qi} + \left(\frac{\frac{i \times n}{4} - C f_{bi}}{f_{qi}} \right) \times w, \quad i = 1, 2, 3$$

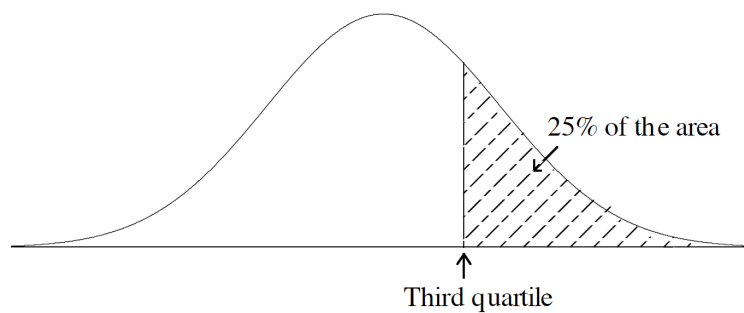
Where $i = 1, 2, 3$ and refers to the quartile number,

L_{qi} : Lower class boundary of the class counting the quartile

$C f_{bi}$: Cumulative frequency before the Q_i class

f_{qi} : The frequency of the Q_i class

w : Class size of the Q_i class.



2.2.2 DECILES

The values of the variable that divide the frequency of the distribution into ten equal parts are known as deciles and are denoted by $D_1, D_2, \dots, D_9$. the fifth deciles is the median ($D_5 = Q_2 = \tilde{X}$).

For ungrouped data, the distribution is first arranged in ascending order of magnitude. Then

$$\begin{aligned} D_1 : & \quad \left(\frac{n+1}{10}\right)^{th} \text{ member of the distribution} \\ & \quad \vdots \\ D_5 : & \quad 5 \times \left(\frac{n+1}{10}\right)^{th} \text{ member of the distribution} \\ & \quad \vdots \\ D_9 : & \quad 9 \times \left(\frac{n+1}{10}\right)^{th} \text{ member of the distribution} \end{aligned}$$

if for $i = 1, \dots, 9$; $i \times \left(\frac{n+1}{10}\right) \notin \mathbb{N}$, then $D_i = \frac{x_{(k)} + x_{(k+1)}}{2}$ where k is the first integer before $i \times \left(\frac{n+1}{10}\right)$, $x_{(k)}$ and $x_{(k+1)}$ are the $(k)^{th}$ and $(k+1)^{th}$ ordered observations.

For a grouped data

$$D_i = L_{Di} + \left(\frac{\frac{i \times n}{10} - Cf_{bi}}{f_{Di}} \right) \times w, \quad i = 1, \dots, 9$$

Where

- L_{Di} : Lower class boundary of the class counting the decile
- Cf_{bi} : Cumulative frequency before the D_i class
- f_{Di} : The frequency of the D_i class
- w : Class size of the D_i class.

2.2.3 PERCENTILE

The values of the variable that divide the frequency of the distribution into hundred equal parts are known as percentiles and are generally denoted by $P_1, \dots, P_{99}$. The fiftieth percentile is the median ($P_{50} = D_5 = Q_2 = \tilde{X}$).

For ungrouped data, the distribution is first arranged in ascending order of magnitude. Then

$$\begin{aligned} P_1 &: \left(\frac{n+1}{100}\right)^{th} \text{ member of the distribution} \\ &\vdots \\ P_{50} &: 50 \times \left(\frac{n+1}{100}\right)^{th} \text{ member of the distribution} \\ &\vdots \\ P_{99} &: 99 \times \left(\frac{n+1}{100}\right)^{th} \text{ member of the distribution} \end{aligned}$$

if for $i = 1, \dots, 99$; $i \times \left(\frac{n+1}{100}\right) \notin \mathbb{N}$, then $P_i = \frac{x_{(k)} + x_{(k+1)}}{2}$ where k is the first integer before $i \times \left(\frac{n+1}{100}\right)$, $x_{(k)}$ and $x_{(k+1)}$ are the $(k)^{th}$ and $(k+1)^{th}$ ordered observations.

For a grouped data

$$P_i = L_{P_i} + \left(\frac{\frac{i \times n}{100} - C f_{bi}}{f_{P_i}} \right) \times w, \quad i = 1, \dots, 99$$

Where

- L_{P_i} : Lower class boundary of the class counting the percentile
- $C f_{bi}$: Cumulative frequency before the P_i class
- f_{P_i} : The frequency of the P_i class
- w : Class size of the P_i class.

Example. For the table below, find by calculation (using appropriate expression)

- (i) Lower quartile, Q_1

(ii) Upper Quartile, Q_3

(iii) 6th Deciles, D_6

(iv) 45th percentile P_{45}

of the following distribution

Marks	20 – 29	30 – 39	40 – 49	50 – 59	60 – 69	70 – 79	80 – 89	90 – 99
Frequency	8	10	14	26	20	16	4	2
Cf	8	18	32	58	78	94	98	100
			↑	↑	↑			
			Q_1	P_{45}	Q_3, D_6			

$$(i) \text{ Lower quartile, } Q_1 = L_{q1} + \left(\frac{\frac{100}{4} - Cf_{b1}}{f_{q1}} \right) \times w = 40 + \left(\frac{25-18}{14} \right) \times 10 = 45$$

$$(ii) \text{ Upper Quartile, } Q_3 = L_{q3} + \left(\frac{\frac{3 \times 100}{4} - Cf_{b3}}{f_{q3}} \right) \times w = 60 + \left(\frac{75-58}{20} \right) \times 10 = 68.5$$

$$(iii) \text{ 6th Deciles, } D_6 = L_{D6} + \left(\frac{\frac{6 \times 100}{10} - Cf_{b6}}{f_{D6}} \right) \times w = 60 + \left(\frac{60-58}{20} \right) \times 10 = 61$$

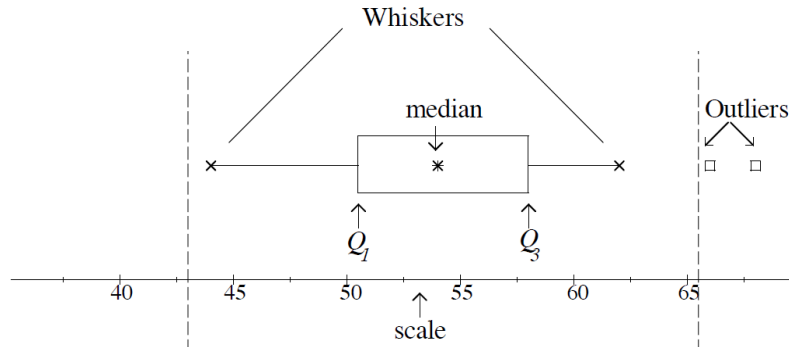
$$(iv) \text{ 45th percentile } P_{45} = L_{P45} + \left(\frac{\frac{45 \times 100}{100} - Cf_{b45}}{f_{P45}} \right) \times w = 50 + \left(\frac{45-32}{26} \right) \times 10 = 55$$

2.2.4 The Box-plot

The box-plot is another way of representing a data set graphically. It is constructed using the quartiles, and gives a good indication of the spread of the data set and its symmetry (or lack of symmetry). It is a very useful method for comparing two or more data sets.

The box-plot consists of a scale, a box drawn between the first and third quartile, the median placed within the box, whiskers on both sides of the box and outliers (if any).

The two dashed vertical lines in the figure are the lower and upper outlier thresholds and are not normally included in a box-plot.



The following data set was used to construct a box-plot :

57 46 61 66 48 59 55 56 60 49 44 53 68 57 55 54 49 50 52 54 62 59 51 52 53 54 47 53

Constructing a Box-plot

Step 1: Order the data and calculate the quartiles.

44 46 47 48 49 49 50

51 52 52 53 53 53 54

54 54 55 55 56 57 57

59 59 60 61 62 66 68

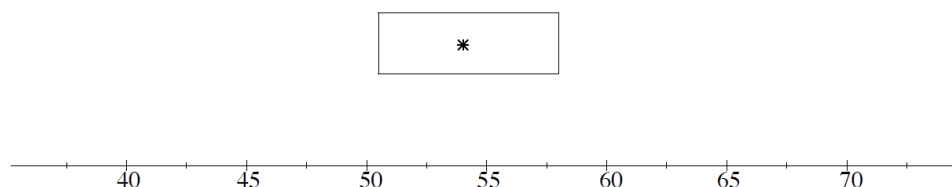
Now we calculate the median, the first quartile and the third quartile.

For these data, median = 54, the first quartile = 50.5 and the third quartile = 58.

With this information we can begin to construct the box-plot.

Step 2: Draw the scale and mark on the quartiles.

Mark the median at the correct place above the scale with a asterix, draw a box around this asterix with the left hand side of the box at the first quartile, 50.5, and the right hand side of the box at the third quartile, 58. This is illustrated in next figure



Step 3: Calculate the interquartile range and determine the position of the outlier thresholds.

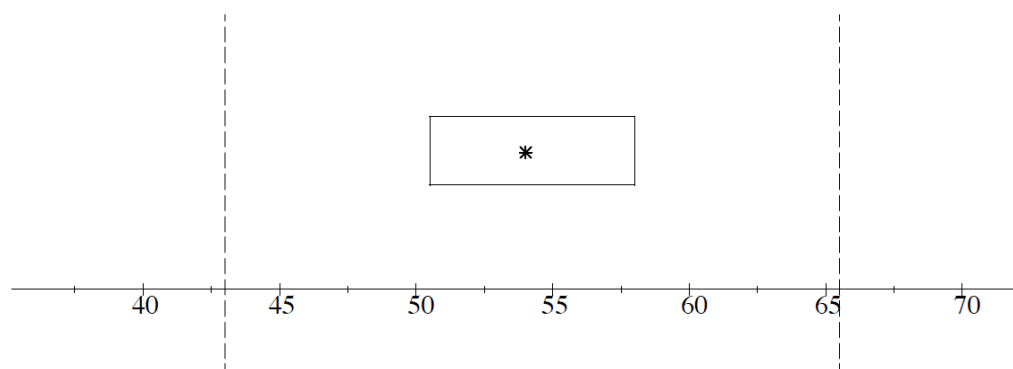
$$\text{Interquartile range} = \text{third quartile} - \text{first quartile} = 58 - 50.5 = 7.5.$$

The position of the lower outlier threshold is found by subtracting the interquartile range from the first quartile, $50.5 - 7.5 = 43$.

The position of the upper outlier threshold is found by adding the interquartile range to the third quartile, $58 + 7.5 = 65.5$.

(Some texts add or subtract $1.5 \times \text{interquartile range}$.)

We now add the outlier thresholds to our diagram. This is illustrated in the figure below.

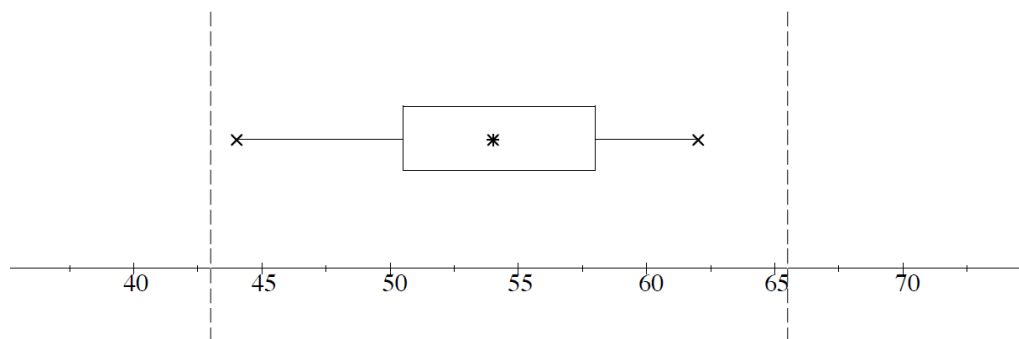


Step 4: Use the outlier thresholds to draw the whiskers.

To draw the left hand whisker, we need the smallest data value that lies inside the outlier thresholds. In this example, it is the value 44. This is drawn on our diagram with a small cross level with the asterix. A horizontal line is now drawn to the left hand side of the box.

To draw the right hand whisker, we find the largest data value that lies inside the outlier thresholds. In this example, the value is 62. This is drawn on the right hand side of the box with a small cross and connected to the box by a horizontal line.

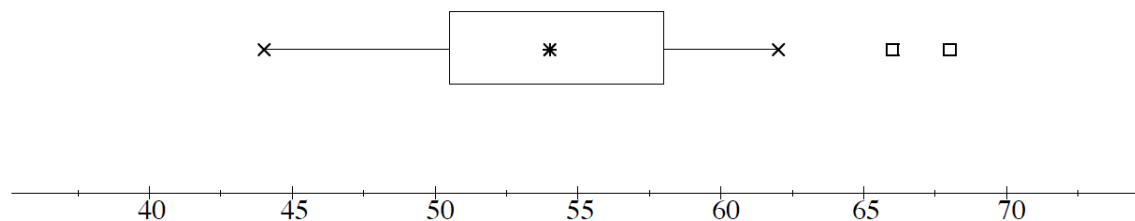
This is illustrated in the next figure.



Step 5: Determine the outliers and remove the outlier thresholds.

Values (if any) that lie outside the outlier thresholds are called outliers. In this example, 66 and 68 are outliers. These are placed on the diagram using a small square or circle. Finally, the outlier thresholds are removed.

The completed box-plot is illustrated in following figure.



2.3 MEASURES OF DISPERSION

Dispersion or variation is degree of scatter or variation of individual value of a variable about the central value such as the median or the mean. These include range, mean deviation, semi-interquartile range, variance, standard deviation and coefficient of variation.

2.3.1 THE RANGE

This is the simplest method of measuring dispersions. It is the difference between the largest and the smallest value in a set of data. It is commonly used in statistical quality control. However, the range may fail to discriminate if the distributions are of different types.

$$Range = L - S$$

2.3.2 SEMI – INTERQUARTILE RANGE

This is the half of the difference between the first (lower) and third quartiles (upper).

It is good measure of spread for midrange and the quartiles.

$$S.I.R = \frac{Q_3 - Q_1}{2}$$

2.3.3 THE MEAN/ABSOLUTE DEVIATION

Mean deviation is the mean absolute deviation from the centre. A measure of the center could be the arithmetic mean or median.

Given a set $x_1, \dots, x_n$, the mean deviation from the arithmetic mean is defined by:

$$AD = \frac{1}{n} \sum_{i=1}^n |x_i - \bar{X}|$$

In a grouped data

$$AD = \frac{1}{\sum_{i=1}^n f_i} \sum_{i=1}^n f_i |x_i - \bar{X}|$$

Example. Below is the average of 6 heads household randomly selected from a country. 47, 45, 56, 60, 41, 54 .Find the (i) Range, (ii) Mean, (iii) Mean deviation from the mean.

The range of the data is $R = 60 - 41 = 19$, and the mean is $\bar{X} = 50.5$. The absolute

deviation from the mean is

$$\begin{aligned}
 AD &= \frac{1}{n} \sum_{i=1}^n |x_i - \bar{X}| \\
 &= \frac{|47 - 50.5| + |45 - 50.5| + |56 - 50.5| + |60 - 50.5| + |41 - 50.5| + |54 - 50.5|}{6} \\
 &= 6.1667
 \end{aligned}$$

Example. The table below shown the frequency distribution of the scores of 42 students.

the mean deviation from the mean ($\bar{X} = \frac{1}{\sum_{i=1}^n f_i} \sum_{i=1}^n f_i x_i = \frac{1429}{42} = 34.024$) for the data is $AD = \frac{1}{\sum_{i=1}^n f_i} \sum_{i=1}^n f_i |x_i - \bar{X}| = \frac{465.76}{42} = 11.09$.

Scores	midpoint x_i	f_i	$f_i x_i$	$ x - \bar{X} $	$f_i x - \bar{X} $
0-9	4.5	2	9	29.52	59.04
10-19	14.5	5	72.5	19.52	97.6
20-29	24.5	8	196	9.52	76.16
30-39	34.5	12	414	0.48	5.76
40-49	44.5	9	400.5	10.48	94.32
50-59	54.5	5	272.5	20.48	102.4
60-69	64.5	1	64.5	30.48	30.48
		42	1429		465.76

2.3.4 THE STANDARD DEVIATION AND VARIANCE

The standard deviation, usually denoted by the Greek alphabet σ (small signal is for the population) is defined as the "positive square root of the arithmetic mean of the squares of the deviation of the given observation from their arithmetic mean". The variance of a set of observations is defined as "the square of the standard deviation" and is thus

given by σ^2 .

Given $x_1, \dots, x_n$ as a set of observations, then the standard deviation and variance are given by:

STD	For population	For samples
scattered data	$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$	$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{X})^2}$
grouped data	$\sigma = \sqrt{\frac{1}{\sum_{i=1}^N f_i} \sum_{i=1}^N f_i (x_i - \mu)^2}$	$s = \sqrt{\frac{1}{(\sum_{i=1}^n f_i) - 1} \sum_{i=1}^n f_i (x_i - \bar{X})^2}$

VAR	For population	For samples
scattered data	$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$	$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{X})^2$
grouped data	$\sigma^2 = \frac{1}{\sum_{i=1}^N f_i} \sum_{i=1}^N f_i (x_i - \mu)^2$	$s^2 = \frac{1}{(\sum_{i=1}^n f_i) - 1} \sum_{i=1}^n f_i (x_i - \bar{X})^2$

When we compute the variance, it is important to understand the unit of measure and what happens when the differences in the numerator are squared. When we calculate the variance, the unit of measure for the variance will be the one of the studied variable squared. There is a way out of this difficulty. By taking the square root of the population variance, we can transform it to the same unit of measurement used for the original data and this is the main importance of the standard deviation.

The formula for the population mean is $\mu = \sum x/N$. We just changed the symbols for the sample mean; that is, $\bar{X} = \sum x/n$. Unfortunately, the conversion from the population variance to the sample variance is not as direct. It requires a change in the denominator. Instead of substituting n (number in the sample) for N (number in the population), the denominator is $n - 1$.

Why is this change made in the denominator? Although the use of n is logical since x is used to estimate μ , it tends to underestimate the population variance, σ^2 . The use of $(n - 1)$ in the denominator provides the appropriate correction for this tendency.

Because the primary use of sample statistics like s^2 is to estimate population parameters like σ^2 , $(n - 1)$ is preferred to n in defining the sample variance. We will also use this convention when computing the sample standard deviation.

Chebyshev's Theorem

We have stressed that a small standard deviation for a set of values indicates that these values are located close to the mean. Conversely, a large standard deviation reveals that the observations are widely scattered about the mean. The Russian mathematician P. L. Chebyshev (1821–1894) developed a theorem that allows us to determine the minimum proportion of the values that lie within a specified number of standard deviations of the mean. For example, according to Chebyshev's theorem, at least three out of every four, or 75%, of the values must lie between the mean plus two standard deviations and the mean minus two standard deviations. This relationship applies regardless of the shape of the distribution. Further, at least eight of nine values, or 88.9%, will lie between plus three standard deviations and minus three standard deviations of the mean. At least 24 of 25 values, or 96%, will lie between plus and minus five standard deviations of the mean.

CHEBYSHEV'S THEOREM *For any set of observations (sample or population), the proportion of the values that lie within k standard deviations of the mean is at least $1 - 1/k^2$, where k is any value greater than 1.*

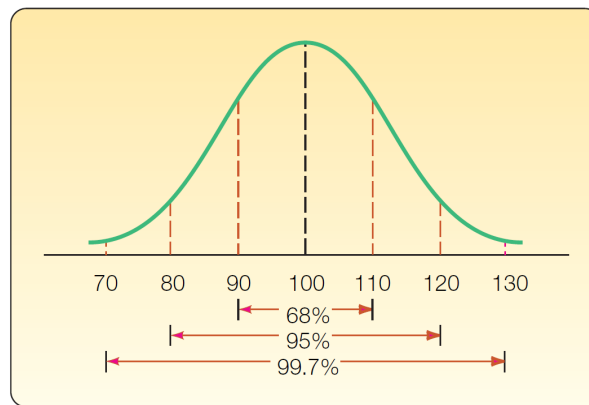
The Empirical Rule

Chebyshev's theorem applies to any set of values; that is, the distribution of values can have any shape. However, for a symmetrical, bell-shaped distribution such as the one in Chart below, we can be more precise in explaining the dispersion about the mean.

These relationships involving the standard deviation and the mean are described by the **Empirical Rule**, sometimes called the **Normal Rule**.

EMPIRICAL RULE *For a symmetrical, bell-shaped frequency distribution, approximately 68% of the observations will lie within plus and minus one standard deviation of the mean; about 95% of the observations will lie within plus and minus two standard deviations of the mean; and practically all (99.7%) will lie within plus and minus three standard deviations of the mean.*

These relationships are portrayed graphically in the below plot for a bell-shaped distribution with a mean of 100 and a standard deviation of 10.



2.3.5 COEFFICIENT OF VARIATION/DISPERSION

This is a dimension less quantity that measures the relative variation between two servers observed in different units. The coefficients of variation are obtained by dividing the standard deviation by the mean and multiply it by 100. Symbolically

$$CV = \frac{\sigma}{\mu} \times 100$$

The distribution with smaller C.V is said to be better.

Example. The data below represents the age of 77 applicants in an achievement test for the post of Botanist in a large company. Compute the

(i) Mean

(ii) Standard deviation

(iii) Coefficient of variation.

Ages(in years)	x_i	f_i	$f_i x_i$	$f_i(x - \bar{X})^2$
50-54	52	1	52	402.40
55-59	57	2	114	453.61
60-64	62	10	620	1012.04
65-69	67	12	804	307.24
70-74	72	18	1296	0.07
75-79	76	25	1925	610.09
80-84	82	9	738	889.23
		77	5549	3674.68

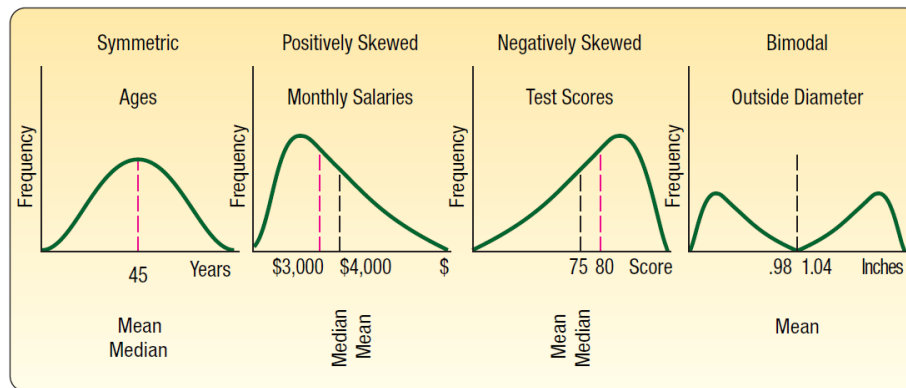
$$(i) \bar{X} = \mu = \frac{5549}{77} = 72.065, (ii) \sigma = \sqrt{\frac{1}{\sum_{i=1}^N f_i} \sum_{i=1}^N f_i (x_i - \mu)^2} = \sqrt{\frac{3674.68}{77}} = 6.9082$$

$$(iii) CV = \frac{\sigma}{\mu} \times 100 = \frac{6.9082}{72.065} \times 100\% = 9.5861\%.$$

2.3.6 SKEWNESS

Another characteristic of a distribution is the shape. There are four shapes commonly observed: symmetric, positively skewed, negatively skewed, and bimodal. In a symmetric distribution the mean and median are equal and the data values are evenly spread around these values. The shape of the distribution below the mean and median is a mirror image of distribution above the mean and median. A distribution of values is

skewed to the right or positively skewed if there is a single peak, but the values extend much farther to the right of the peak than to the left of the peak. In this case, the mean is larger than the median. In a negatively skewed distribution there is a single peak, but the observations extend farther to the left, in the negative direction, than to the right. In a negatively skewed distribution, the mean is smaller than the median. Positively skewed distributions are more common. Salaries often follow this pattern. Think of the salaries of those employed in a small company of about 100 people. The president and a few top executives would have very large salaries relative to the other workers and hence the distribution of salaries would exhibit positive skewness. A bimodal distribution will have two or more peaks. This is often the case when the values are from two or more populations. This information is summarized in the following figure.



There are several formulas in the statistical literature used to calculate skewness. The simplest, developed by Professor Karl Pearson (1857–1936), is based on the difference between the mean and the median.

PEARSON'S COEFFICIENT OF SKEWNESS

$$sk = \frac{3(\bar{X} - \tilde{X})}{s}$$

Using this relationship, the coefficient of skewness can range from -3 up to 3. A value near -3, such as -2.57, indicates considerable negative skewness. A value such as 1.63 indicates moderate positive skewness. A value of 0, which will occur when the mean and median are equal, indicates the distribution is symmetrical and there is no skewness present.

MEASURE OF SKEWNESS LARGELY USED IN SOFTWARES

$$sk = \frac{n}{(n-1)(n-2)} \sum \left(\frac{x - \bar{X}}{s} \right)^3$$

This formula offers an insight into skewness. The right-hand side of the formula is the difference between each value and the mean, divided by the standard deviation. That is the portion $\frac{x - \bar{X}}{s}$ of the formula. This idea is called **standardizing**. We will discuss the idea of standardizing a value in more detail later when we describe the normal probability distribution. At this point, observe that the result is to report the difference between each value and the mean in units of the standard deviation. If this difference is positive, the particular value is larger than the mean; if the value is negative, the standardized quantity is smaller than the mean. When we cube these values, we retain the information on the direction of the difference. Recall that in the formula for the standard deviation we squared the difference between each value and the mean, so that the result was all nonnegative values.

If the set of data values under consideration is symmetric, when we cube the standardized values and sum over all the values, the result would be near zero. If there are several large values, clearly separate from the others, the sum of the cubed differences

would be a large positive value. If there are several small values clearly separate from the others, the sum of the cubed differences will be negative.